

IFS:

(X, d) - complete metric space

$f_1, f_2, \dots, f_N$ contraction maps on (X, d) .

Then $\exists$ a unique compact set $A \subseteq X$
such that $A = \bigcup_{i=1}^N f_i(A)$.

A is called an attractor or invariant set for the IFS.

$F : \mathcal{K}(X) \rightarrow \mathcal{K}(X)$ $\rightarrow$ space of all non-empty compact subsets of X with the Hausdorff metric

$$F(K) = \bigcup_{i=1}^N f_i(K)$$

A is the unique fixed point of F .

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Also, $D(F^n(K), A) \rightarrow 0$ as $n \rightarrow \infty$
for any $K \in \mathcal{K}(X)$.

$$\text{In fact, } D(F(K), A) = D(F(K), F(A)) \leq c D(K, A),$$

$$\text{where } c = \max\{c_i : 1 \leq i \leq N\} < 1$$

$$\therefore D(F^n(K), A) \leq c^n D(K, A) \rightarrow 0 \text{ as } n \rightarrow \infty.$$

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Code space & code map :

Let $\{(X, d); f_1, f_2, \dots, f_N\}$ be an IFS.

Let $\Lambda = \{1, 2, \dots, N\} \rightarrow$ alphabet set

For $w = (i_1, i_2, \dots, i_n) \in \Lambda^n$, we define

$$f_w = f_{i_1} \circ f_{i_2} \circ \dots \circ f_{i_n}$$

Since, $K = \bigcup_{i=1}^N f_i(K)$,

$$K = \bigcup_{w \in \Lambda^n} f_w(K)$$

Let $w = (i_1, i_2, \dots, i_n, \dots) \in \Lambda^\mathbb{N}$

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For $x \in X$ and $w = (i_1, i_2, \dots) \in \Lambda^\mathbb{N}$

$$x_n = f_{i_1} \circ f_{i_2} \circ \dots \circ f_{i_n}(x)$$

Then $\{x_n\}_{n=1}^\infty$ is a Cauchy seq.

(If $n \geq m$, $d(x_n, x_m) = d(f_{i_1} \circ \dots \circ f_{i_m}(f_{i_{m+1}} \circ \dots \circ f_{i_n}(x)), f_{i_1} \circ \dots \circ f_{i_m}(x))$

$$\leq \sum_{k=m+1}^n d(f_{i_k} \circ \dots \circ f_{i_n}(x), x) \rightarrow 0 \text{ as } n \rightarrow \infty$$

Since (X, d) is complete,

$\{x_n\}$ converges to some point in X ,

say $\pi(x, w)$

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Suppose $x, y \in X$

Claim: $\pi(x, w) = \pi(y, w)$ for any $w \in \Delta^N$.

$$\pi(x, w) = \lim_{n \rightarrow \infty} f_{i_1} \circ \dots \circ f_{i_n}(x)$$

$$\pi(y, w) = \lim_{n \rightarrow \infty} f_{i_1} \circ \dots \circ f_{i_n}(y)$$

Now, $d(f_{i_1} \circ \dots \circ f_{i_n}(x), f_{i_1} \circ \dots \circ f_{i_n}(y))$
 $\leq c^n d(x, y) \rightarrow 0$ as $n \rightarrow \infty$.

$\Rightarrow \pi(x, w) = \pi(y, w)$.

$\Rightarrow$ Thus we can define a map $\pi: \Delta^N \rightarrow X$
 by $\pi((i_1, i_2, \dots, i_n, -)) = \lim_{n \rightarrow \infty} f_{i_1} \circ \dots \circ f_{i_n}(x)$
 for any $x \in X$.

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Let $J = \{ \pi(w) : w \in \Delta^N \}$

$J = \bigcup_{i=1}^N f_i(J)$

Proof: $f_i(\pi(w)) = f_i\left(\lim_{n \rightarrow \infty} f_{i_1} \circ \dots \circ f_{i_n}(x)\right)$
 $= \lim_{n \rightarrow \infty} f_i \circ f_{i_1} \circ \dots \circ f_{i_n}(x)$
 $(\because f_i \text{ is continuous})$
 $= \pi(iw)$

where if $w = (i_1, i_2, \dots, i_n, -)$,
 $iw = (i, i_1, i_2, \dots, i_n, -)$

$\therefore f_i(J) \subseteq J$ for each i

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Conversely, let $x \in J$.
 Then $x = \pi(w)$ for some $w = (i_1, i_2, \dots) \in \Delta^{\mathbb{N}}$

$$= f_{i_1}(\pi((i_2, i_3, \dots)))$$

$$\in \bigcup_{i=1}^N f_i(J)$$

• Claim: J is nonempty compact set.

$$\pi : \Delta^{\mathbb{N}} \rightarrow X$$

If we take the product topology on $\Delta^{\mathbb{N}}$ (discrete topology on $\Delta = \{1, 2, \dots, N\}$),
 then the map π is continuous.

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Since Δ is compact, $\Delta^{\mathbb{N}}$ is compact
 by the Tychonoff's theorem.

Hence, $J = \pi(\Delta^{\mathbb{N}})$ is compact.

By the uniqueness of attractor,
 J is the attractor for the IFS.

• $\Delta^{\mathbb{N}}$ is called the code space
 • $\pi : \Delta^{\mathbb{N}} \rightarrow X$ is called the code map.

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Self-similar sets

(X, d) — complete metric space
 Assume $f_1, f_2, \dots, f_N$ are similarity maps,
 i.e., $d(f_i(x), f_i(y)) = c_i d(x, y) \quad \forall x, y \in X$

Also, assume that each $c_i \in (0, 1)$.
 Then $\exists$ a unique attractor $A \subseteq X$
 satisfying $A = \bigcup_{i=1}^N f_i(A)$

A is called a self-similar set.

e.g. Cantor set

$$f_1(x) = \frac{x}{3}, \quad f_2(x) = \frac{2}{3} + \frac{x}{3}$$

$$c_1 = c_2 = \frac{1}{3}$$

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Similarity dimension:

There exists a unique $s \in [0, \infty)$ s.t.

$$\sum_{i=1}^N c_i^s = 1$$

This s is called the "similarity dimension"
 of the self-similar set A .

(Let $\varphi : (0, \infty) \rightarrow \mathbb{R}$ be defined by
 $\varphi(s) = \sum_{i=1}^N c_i^s$)

Then φ is continuous.

φ is strictly decreasing.

$$\varphi(0) = N \geq 1$$

$$\lim_{s \rightarrow \infty} \varphi(s) = 0$$

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e.g. For the Cantor set, C , the similarity dimension is given by

$$\left(\frac{1}{3}\right)^s + \left(\frac{1}{3}\right)^s = 1$$

$$\Rightarrow \frac{2}{3^s} = 1 \Rightarrow \boxed{s = \frac{\ln 2}{\ln 3}}$$

For Sierpinski gasket,

$$c_1 = c_2 = c_3 = \frac{1}{2}$$

$$3 \left(\frac{1}{2}\right)^s = 1 \Rightarrow 2^s = 3 \Rightarrow \boxed{s = \frac{\ln 3}{\ln 2}}$$

For the Koch curve, $c_1 = c_2 = c_3 = c_4 = \frac{1}{3}$

$$4 \left(\frac{1}{3}\right)^s = 1 \Rightarrow \boxed{s = \frac{\ln 4}{\ln 3}}$$

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Separation conditions:

Let $\{X, d\}; f_1, f_2, \dots, f_N\}$ be an IFS.

Let A be the attractor for the IFS.

Strong separation condition (SSC)

$$f_i(A) \cap f_j(A) = \emptyset \text{ for } i \neq j.$$

Open set condition (OSC)

$\exists$ a nonempty open set $V \subseteq X$
 st. $f_i(V) \cap f_j(V) = \emptyset$ for $i \neq j$
 and $\bigcup_{i=1}^N f_i(V) \subseteq V$.

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Strong open set condition (SOSC):
 $\exists$ a nonempty set $V \subseteq X$ st.
 $f_i(V) \cap f_j(V) = \emptyset$, $\bigcup_{i=1}^N f_i(V) \subseteq V$
 and $V \cap A \neq \emptyset$, where A is the attractor.

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